Learning and Stabilizing Memories in Noisy Recurrent Spiking Neural Networks

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Introduction

How does the brain encode and stabilize a new memory or activity pattern in a noisy environment? Although current prevailing theories suggest that memories are encoded in patterns of synaptic weights within brain networks , the mechanisms by which these patterns are stabilized and reliably accessed are not well understood. Individual synapses and neurons are unreliable, and their weights and responses can change over time.

Recent studies suggest that network activity patterns could be stabilized in rate based neural networks via complex plasticity mechanisms. However, it is still unclear whether the same mechanisms could produce stable memories in spiking neural networks and whether these learning rules are biologically plausible.

In our work, we are developing a biologically inspired spiking neural network that can learn and store multiple memories despite the presence of noise.

Methodology

Izhikevich Model of Spiking Neurons

$$\frac{dv}{dt} = 0.04v^2 + 5v + 140 - u + I$$

Neuronal Input Current

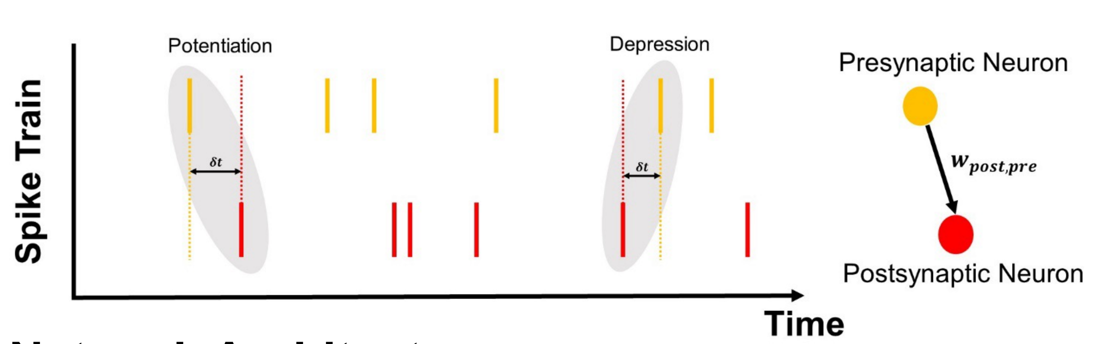
$$\frac{du}{dt} = a(bv - u)$$

$$if\ v \geq 30\ mV, then \begin{cases} v \to c \\ u \to u + d \end{cases}$$

Hebbian Spike Time Dependent Plasticity Rule

Potentiation
$$\begin{cases} w = w + w_p \\ w_p = c_p w \log\left(\frac{w_{max}}{w}\right) e^{-\delta t/\tau_{STDP}} \end{cases}$$

Depression
$$\begin{cases} w = w - w_d \\ w_d = c_d e^{-\delta t/\tau_{STDP}} \end{cases}$$



Network Architecture

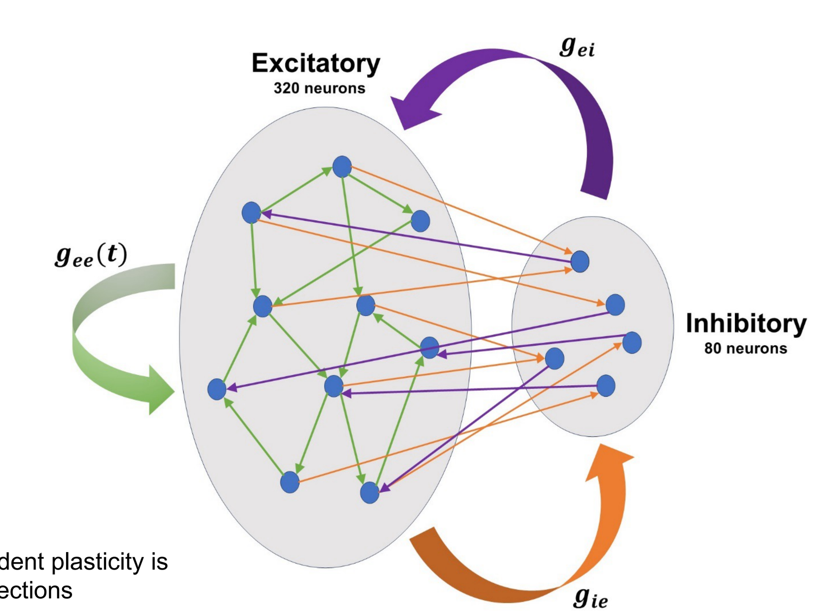


Fig 1. spike time dependent plasticity is only active for g_{ee} connections

Stimulation and Noise Protocol

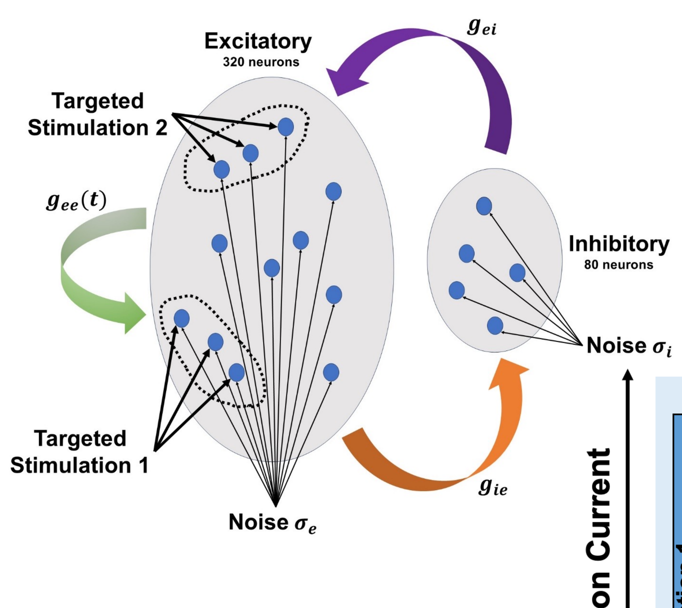
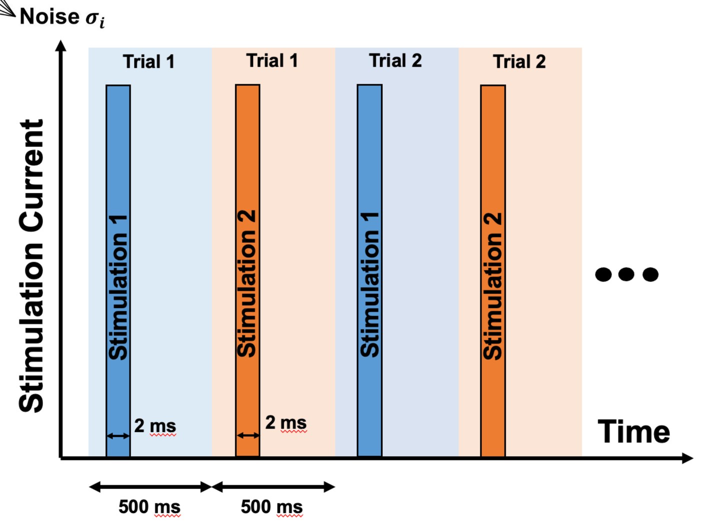


Fig 2. Noise is white noise and added with same amplitude to all neurons. Stimulation subpopulation is 50 randomly chosen cells but two subpopulations are different.

Fig 3. Stimulations are brief but strong currents injected to the targeted subpopulations and last 2 ms.



Results

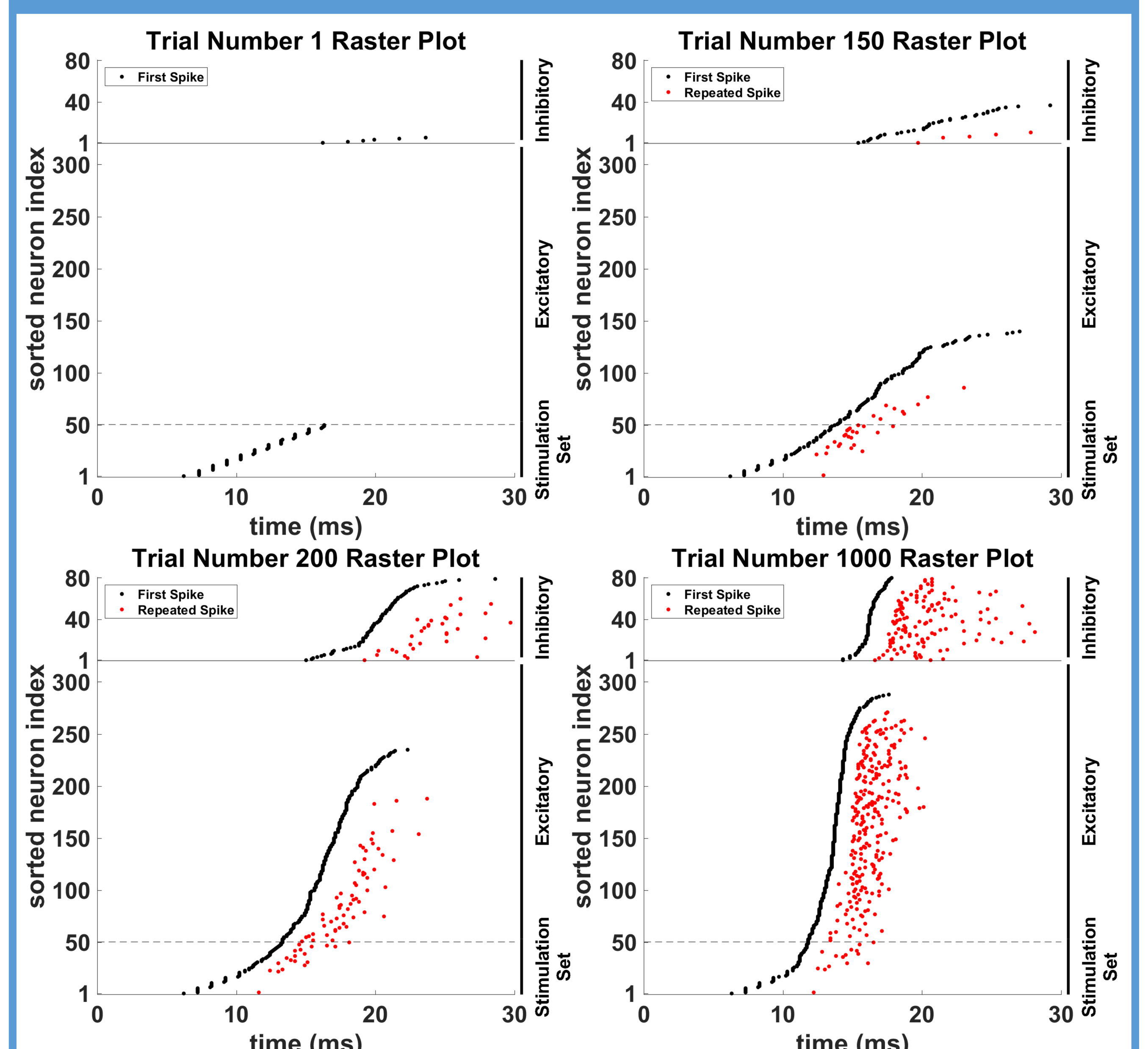


Fig 4. In the top four plots, raster plots of the network response to the stimulation 2 are shown for different consecutive trials. The trial 1000 is the stabilized responses of network to the stimulation 2.

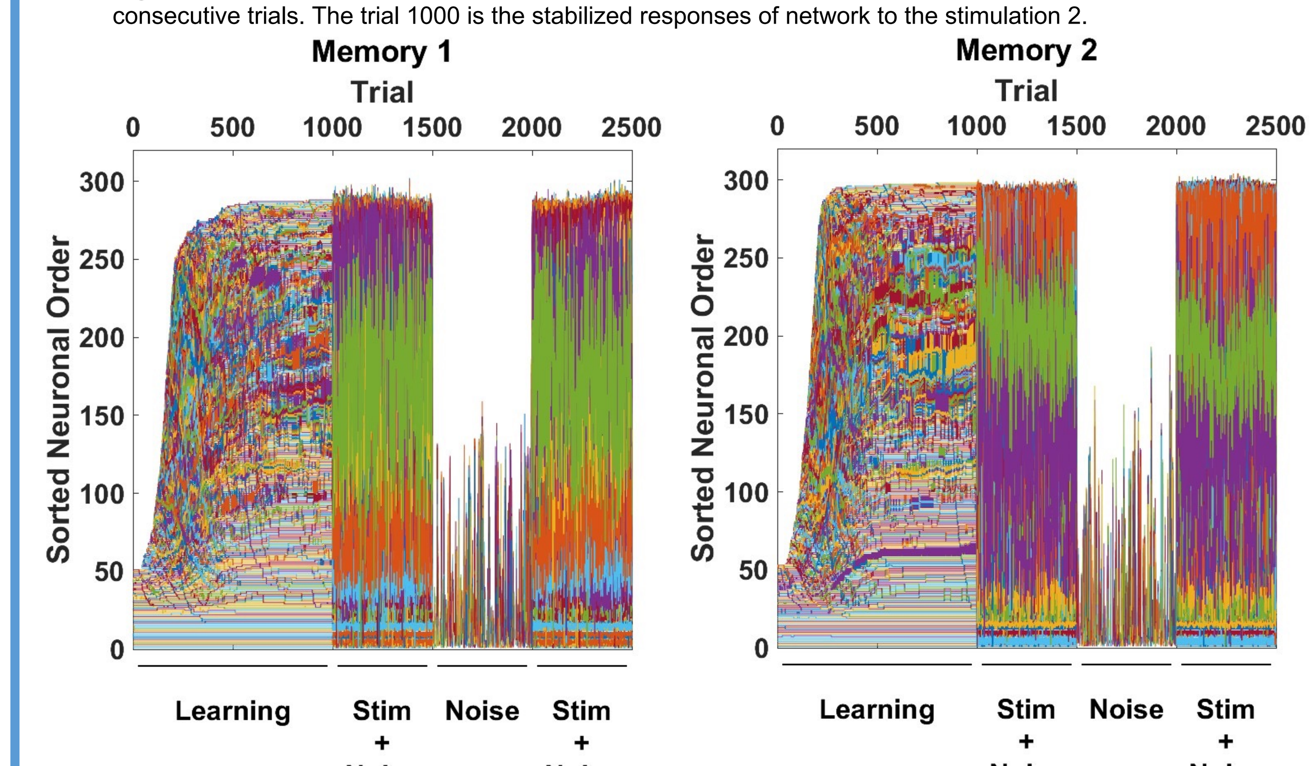


Fig 5. Neuronal indices sorted by time to first spike in each trial. In learning phase network initially has only 50 active cells which occupy first 50 indices and as trials go further, new neurons join this ranking.

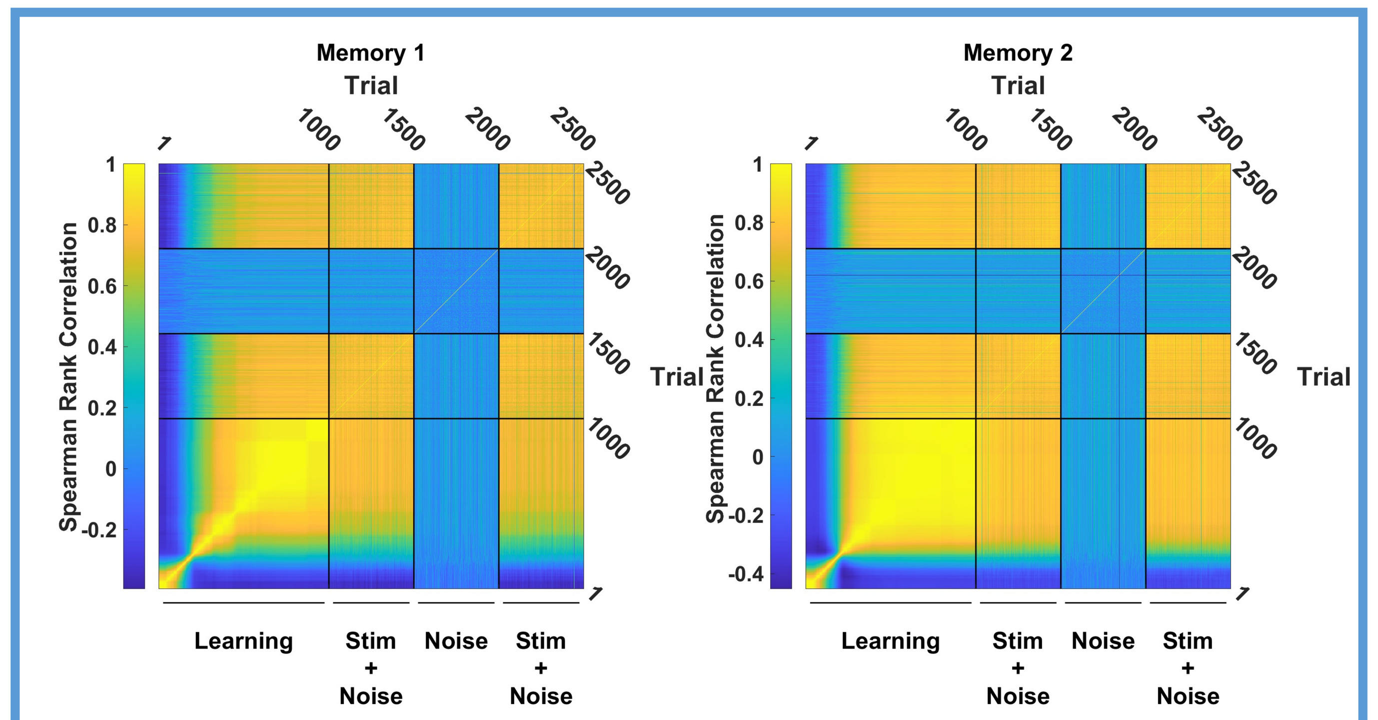


Fig 6. Measure of information stability for two different stimulations. Spearman rank correlation matrix between different time to first spike orders of trials through different phases (Learning, Stimulation + Noise, Noise, Stimulation + Noise).

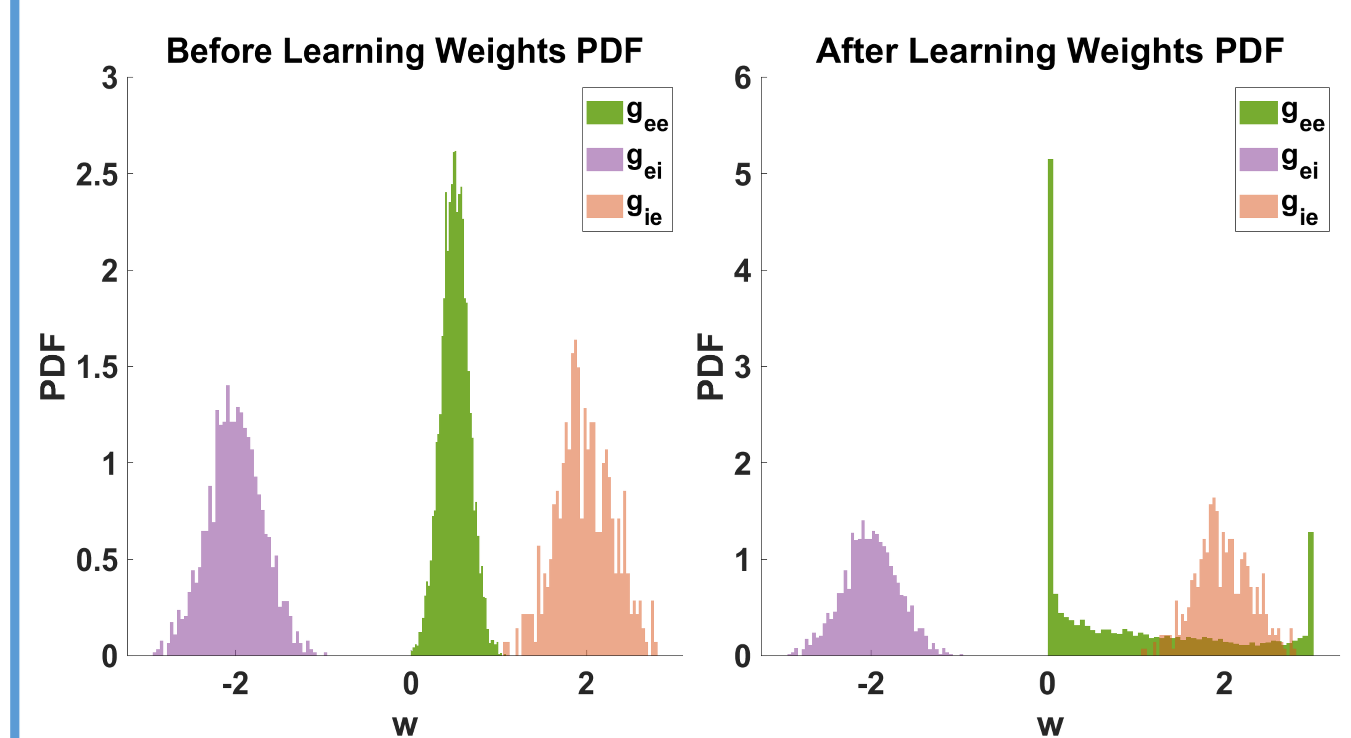


Fig 7. STDP rule evolving synaptic weights (just g_{ee}) in time. Left) Histogram of synaptic weights before learning. Right) Histogram of synaptic weights after learning phase.

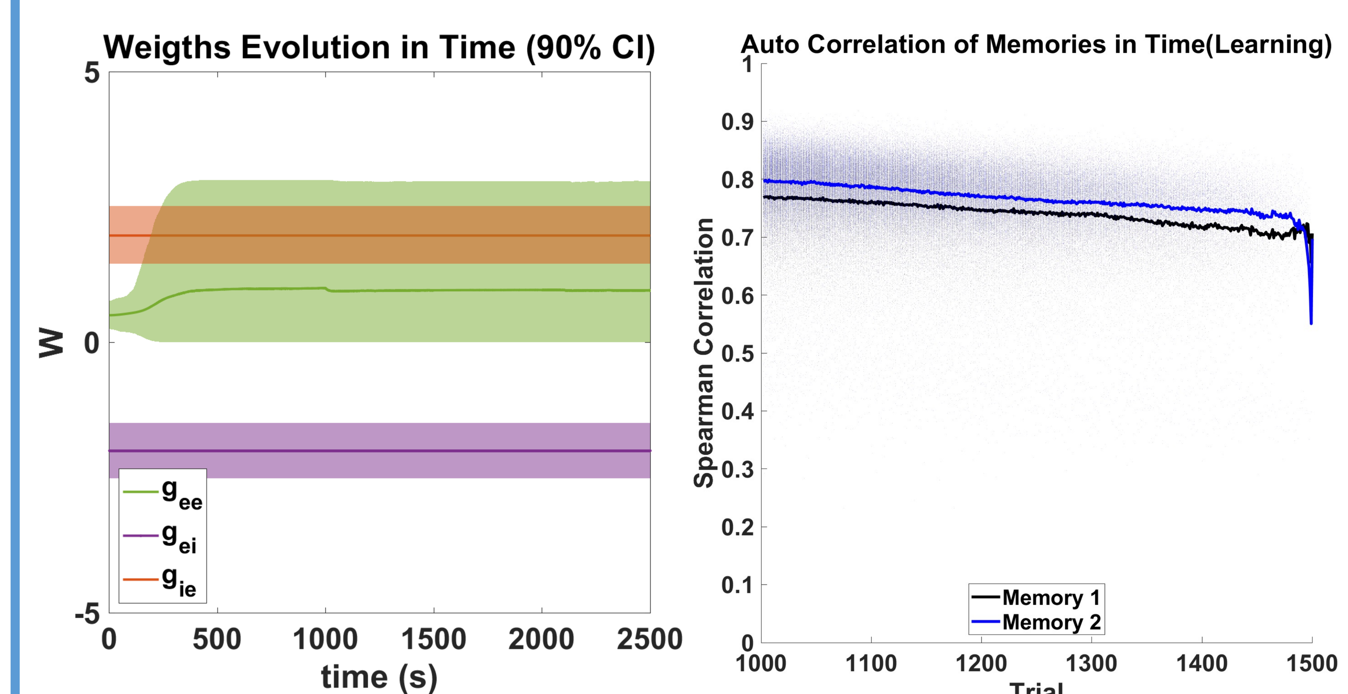


Fig 8. Left) Evolution of excitatory to excitatory synaptic weights in time driven by STDP rule. Right) Auto correlation of first time to spike orderings in stimulation + noise phase which shows a slow decay in similarity of orders.

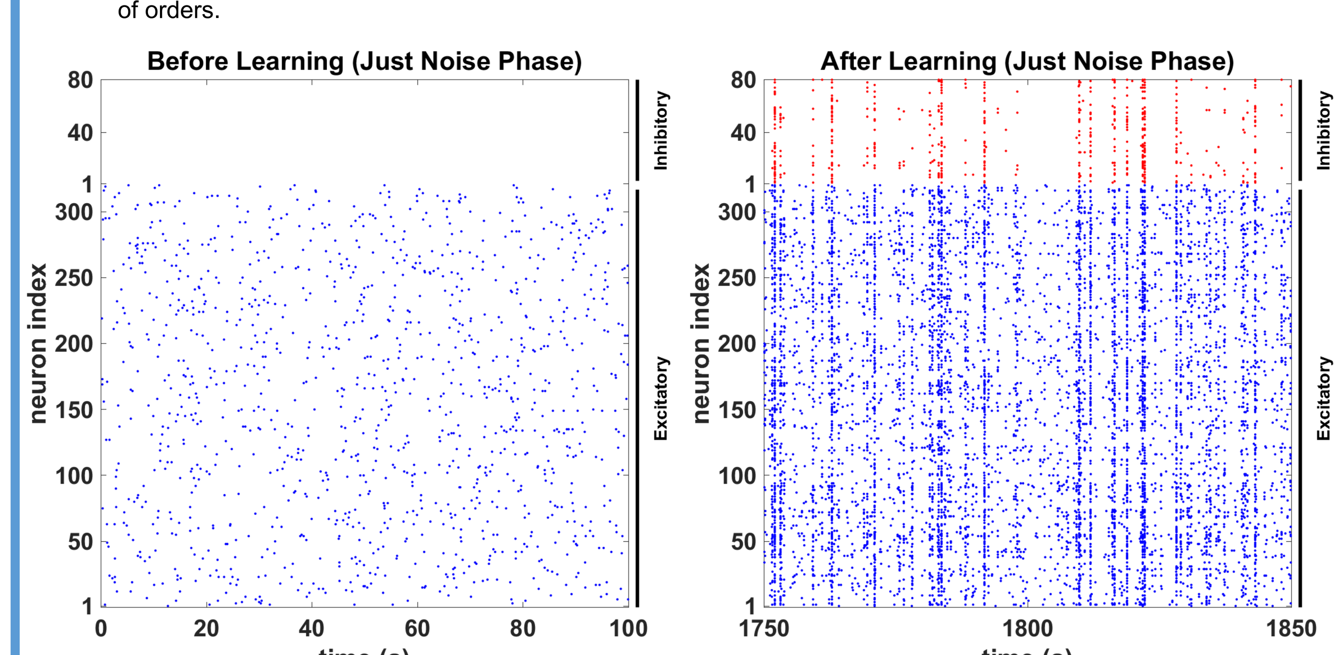


Fig 9. Raster plots of network activity driven just by noise. Left) Noise driven activity of the network before stimulation and learning. Right) Noise driven activity of the network after learning and stimulation.

Conclusions

The results of this research demonstrate that it is feasible to construct a biologically inspired spiking neural network capable of storing memories in a manner similar to classical Hopfield networks. Our findings indicate that such networks can stabilize and maintain memory patterns despite the presence of noise, which is a significant challenge in neural network models.

Future Works

- **Decoder Development:** Build a decoder population of neurons to read the current network activity and identify active memories online.
- **Heterogeneity:** Introduce varied parameters in Izhikevich model for more realistic modeling and more degrees of freedom in the system.
- **Memory Capacity:** Calculate the network's memory storage and retrieval limits.
- **Feature Analysis:** Examine additional features beyond time to first spike ordering of learned activity which could be timing of other neuronal spikes after the first one.
- **Plasticity Mechanism:** incorporate other plasticity mechanisms like synaptic scaling and more detailed STDP rules. Also applying plasticity to other two g_{ei} and g_{ie} connections.

References

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